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Predicting local lithium-plating onset with upscaled models

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## Abstract

Lithium plating is governed by microscale transport heterogeneity, yet conventional macroscale battery models resolve only spatially-averaged quantities. In this work, we extend the dual-continuum (DC) and Doyle–Fuller–Newman (DFN) frameworks to predict local lithium-plating onset within negative-electrode microstructures while retaining the computational efficiency of upscaled approaches. A closure problem for electrolyte transport is introduced to link averaged electrolyte variables to underlying geometric features, permitting reconstruction of local lithium-plating overpotential fields from an upscaled solution. Comparison with direct numerical simulations (DNS) demonstrates that the method accurately identifies regions of highest plating risk. Beyond pinpointing *where* lithium plating is most likely, the closure is further used to propose a correction to the DC and DFN models, improving predictions of *when* plating first becomes thermodynamically favourable. We envisage two primary applications for this framework: (i) in electrode design, to efficiently evaluate candidate microstructures without resorting to computationally intensive DNS; (ii) to improve upscaled models, by enhancing their estimation of when lithium plating becomes favourable. This framework is readily extendable to study other degradation mechanisms driven by microscale transport heterogeneity. **Finally, the closure problem opens a pathway to understanding the geometrical features that drive lithium-plating risk.**

# 1 Introduction

The performance and safety of lithium-ion batteries are critically dependent on degradation mechanisms in the negative electrode [1–3]. Lithium plating is amongst the most notorious of these, and is aggravated during fast charging and low-temperature operation. It can lead to capacity fade, increased cell resistance, and the formation of dendrites that pose a risk of short circuit and thermal runaway [4, 5]. Experimentally, this phenomenon can be studied using techniques such as magnetic resonance imaging (MRI) [6, 7], which provides non-destructive, spatially resolved insight into lithium deposition inside working batteries. Whilst such experiments provide indispensable information, accurately modelling lithium-plating onset is crucial for streamlining cell design and improving battery management systems [8, 9]. The lithium plating reaction [10],



is thermodynamically favourable when the overpotential driving the reaction,  $\eta_{\text{pl}}$ , becomes negative [10–12]. Due to the thin nature of electrodes (often less than  $100 \mu\text{m}$ ), experimentally measuring local potential is extremely challenging [13]. Microscale simulations [14–17] of lithium-ion cells – where transport equations are resolved on a mesh representing the electrode microstructure – provide a means to resolve local potential at all points within the electrode, giving valuable insights into the relationship between microstructure and degradation. In particular, the onset of lithium plating can be predicted using such simulations, by observing when and where  $\eta_{\text{pl}}$  becomes negative.

Whilst microscale simulations offer an abundance of detail, they require enormous computational resource due to the complexity of the mesh required to describe the electrode microstructure. For this reason, upscaled models have become extremely popular, as they offer a physics-based approach to simulate a battery (dis)charge within a fraction of a second on a laptop. Amongst such approaches, the Doyle–Fuller–Newman (DFN) model is likely the most prevalent (also known as the Newman or P2D model). Recently, we published a novel upscaled model of a lithium-ion battery [18–20]. Our model falls into the dual-continuum (DC) category of upscaled formulations [21], a framework that has recently seen renewed attention for lithium-ion battery modelling following the work of Yang and Tartakovsky [22]. The DC model we present captures the transport characteristics of the electrode microstructure via a closure problem, thus avoiding the DFN model’s assumption that active-material particles are isolated spheres.

Teel et al. [8] recently demonstrated that local lithium plating in negative electrode structures may be overlooked by upscaled models. They generated a range of electrodes possessing the same macroscale properties – namely their dimensions, porosity, and mean particle radius – but different microstructures. Inevitably, the electrodes exhibited slightly different tortuosities, as it is not possible to enforce identical values. They simulated a galvanostatic charge at various C-rates, and recorded the time at which lithium plating became favourable at any point in the electrode. At higher C-rates (2 C), the onset of lithium plating varied by up to 150 s for the different electrodes. Therefore, the authors conclude that microscale simulations are crucial for such information, since plating may become favourable at certain points in the microstructure even when the average value of  $\eta_{\text{pl}}$  may still be above 0 V. They argue that upscaled models (such as the DFN) would not be able to capture this, since they inherently solve for the averaged value of  $\eta_{\text{pl}}$ .

In this work, we demonstrate that it is possible to capture local lithium-plating onset with an upscaled model by using a closure problem – this provides a strong link between averaged variables and the microscale variables in the real electrode geometry. Using the volume-averaging technique (VAT), we develop a closure that facilitates *reconstruction* of  $\eta_{\text{pl}}$  within the microstructure when coupled with upscaled model solutions; this differs from our previous work [18, 19] in that it describes electrolyte transport rather than active-material transport. We then validate the theory developed with a study analogous to Teel’s work, using the closure to predict areas in the microstructure at risk of plating, and also to predict the time at which lithium plating first becomes thermodynamically favourable. Finally, since the separator-electrode interface is of paramount importance when studying lithium-plating, we extend the validation to include a case where the real separator structure is modelled.

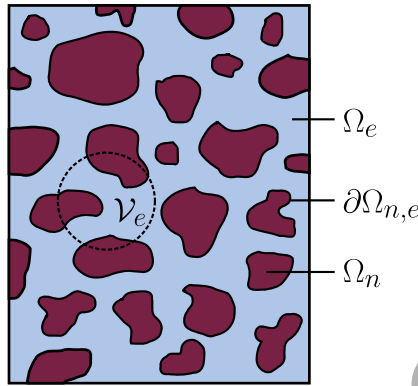


Figure 1: A schematic of the electrolyte domain ( $\Omega_e$ ), the AM domain ( $\Omega_n$ ), the electrolyte-AM interface ( $\partial\Omega_{n,e}$ ), and an example averaging volume ( $\mathcal{V}_e$ ).

## 2 Theory

### 2.1 Problem Description

We consider a negative electrode composed of two phases, electrolyte and active material (AM), that are at a constant and uniform temperature – a schematic is shown in Figure 1. The electrolyte domain is denoted as  $\Omega_e$ , whilst that of the AM is  $\Omega_n$ . The AM-electrolyte interface is defined as  $\partial\Omega_{n,e}$ , and the domain encompassing both phases is called  $\Omega = \Omega_e \cup \Omega_n$ .

The reference used to evaluate upscaled models throughout this work is the direct numerical solutions (DNS) to the microscale model, which is provided in the Supplementary Material (SM). In the microscale model, the overpotential for the lithium-plating reaction is

$$\eta_{\text{pl}} = \phi_n - \phi_e, \quad (2)$$

where  $\phi_n$  is the negative electrode AM potential and  $\phi_e$  is the electrolyte potential – all potentials are defined relative to lithium metal. As an aside, Teel [8] refers to  $\eta_{\text{pl}}$  as the ‘solid anode potential’ – this is not to be confused with  $\phi_n$ . Rather, this is an alternative definition of the overpotential referring to the electrode potential relative to a reference electrode adjacent to the surface; see Newman’s chapter on electrode kinetics for a discussion on different overpotential definitions [23]. Returning to Equation 2, the minimum plating overpotential in the microscale model is,

$$\eta_{\text{pl},\text{min}} = \min(\phi_n - \phi_e). \quad (3)$$

In the microscale model,  $\eta_{\text{pl}}$  is defined at all points on  $\partial\Omega_{n,e}$ . This is not the case for upscaled models, since they solve for spatially-smoothed variables, which are defined using an average operator,

$$\langle u \rangle^\gamma(\mathbf{x}, t) \equiv \frac{1}{V_\gamma(\mathbf{x})} \int_{\zeta \in \mathcal{V}_\gamma(\mathbf{x})} u(\zeta, t) dV_\zeta, \quad (4)$$

where  $u$  is a microscale variable,  $\langle u \rangle^\gamma$  is the phase-intrinsic average of  $u$ ,  $\gamma \in \{e, n\}$  is the phase indicator,  $\mathcal{V}_\gamma(\mathbf{x}) \subset \Omega_\gamma$  is an averaging set centered at  $\mathbf{x} \in \Omega$ , and  $V_\gamma(\mathbf{x}) \equiv \int_{\zeta \in \mathcal{V}_\gamma(\mathbf{x})} dV_\zeta$ . Note that we use a simplified version of the averaging operator, and that the general definition employs a spatial convolution [21, 24–26]. Since the upscaled models only resolve spatially smoothed variables, the lithium-plating overpotential that they will predict is

$$\eta_{\text{pl}}^0 = \langle \phi_n \rangle^n - \langle \phi_e \rangle^e, \quad (5)$$

and the minimum value of is therefore,

$$\eta_{\text{pl},\text{min}}^0 = \min(\langle \phi_n \rangle^n - \langle \phi_e \rangle^e). \quad (6)$$

Our goal is to estimate  $\eta_{\text{pl}}$  across the microstructure without requiring solutions to the microscale model. To do so we decompose the microscale variables into an average contribution and a deviation from the average (which is standard in the VAT [27]),

$$\phi_n = \langle \phi_n \rangle^n + \tilde{\phi}_n, \quad (7a)$$

$$\phi_e = \langle \phi_e \rangle^e + \tilde{\phi}_e. \quad (7b)$$

Inserting these into the microscale definition of plating overpotential (Equation 2),

$$\eta_{\text{pl}} = \langle \phi_n \rangle^n + \tilde{\phi}_n - \langle \phi_e \rangle^e - \tilde{\phi}_e. \quad (8)$$

The negative electrode conductivity,  $\sigma_n$ , is typically two orders of magnitude larger than the electrolyte conductivity,  $\sigma_e$  [28, 29], and thus Equation 8 can be simplified by assuming that  $\tilde{\phi}_n \ll \tilde{\phi}_e$ . For the cases studied, we verified the validity of this assumption by comparing the fields of  $\tilde{\phi}_n$  and  $\tilde{\phi}_e$  in the DNS. **One can imagine scenarios that would increase the magnitude of  $\tilde{\phi}_n$ , making this assumption less reliable: e.g. cracks in the AM, or severe contact resistance between particles. For such cases, this assumption could be avoided, which would simply result in a more complex closure.** The expression for  $\eta_{\text{pl}}$  is now

$$\eta_{\text{pl}} = \langle \phi_n \rangle^n - \langle \phi_e \rangle^e - \tilde{\phi}_e. \quad (9)$$

Here, the information that we do not immediately have access to in an upscaled model is  $\tilde{\phi}_e$ .

## 2.2 Volume Averaging

To express  $\tilde{\phi}_e$  as a function of spatially-smoothed variables, thus obtaining ‘closure’ between the microscale and upscaled models, we will apply the VAT to the microscale partial differential equation (PDE) and boundary condition (BC) for  $\phi_e$ . However, since the governing equations for  $\phi_e$  are coupled to those for electrolyte concentration,  $c_e$ , we must also apply the VAT to these equations.

### 2.2.1 Microscale

The microscale PDE and BC on  $\phi_e$  are given by [14, 30],

$$\nabla \cdot \mathbf{i}_e = 0 \quad \text{in } \Omega_e, \quad (10a)$$

$$\mathbf{n} \cdot \mathbf{i}_e = -j_n \quad \text{at } \partial\Omega_{n,e}. \quad (10b)$$

where  $\mathbf{n}$  is the normal vector pointing outwards from the electrolyte, and  $j_n$  is the reaction rate, which is governed by Butler-Volmer kinetics (see SM). The microscale mass conservation equations are

$$\partial_t c_e + \nabla \cdot \mathbf{N}_e = 0 \quad \text{in } \Omega_e, \quad (11a)$$

$$-\mathbf{n} \cdot \mathbf{N}_e = F^{-1} j_n \quad \text{at } \partial\Omega_{n,e}. \quad (11b)$$

The electrolyte current density,  $\mathbf{i}_e$ , and flux,  $\mathbf{N}_e$  are

$$\mathbf{i}_e = -\sigma_e \nabla \phi_e - \sigma_e \theta \nabla \ln c_e, \quad (12a)$$

$$\mathbf{N}_e = -D_e \nabla c_e + \frac{t_+^0}{F} \mathbf{i}_e, \quad (12b)$$

where  $\sigma_e$ ,  $D_e$ , and  $t_+^0$  are the electrolyte conductivity, diffusivity and transference number respectively,

$$\theta = -\frac{2RT}{F} \frac{(1 - t_+^0)}{(1 + \delta_e)}, \quad (13)$$

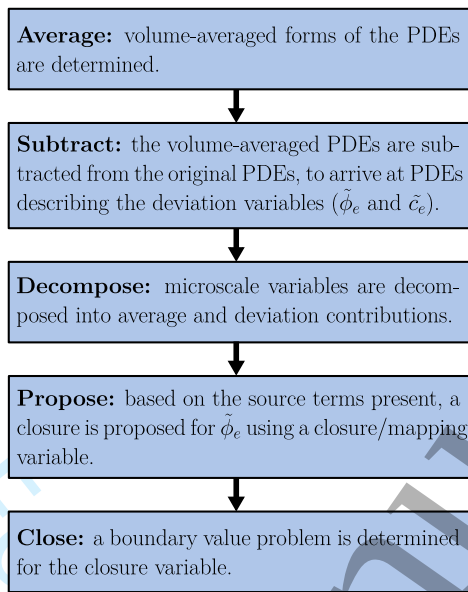


Figure 2: A summary of the volume-averaging process.

in which  $R$  is the gas constant,  $T$  is temperature,  $F$  is the Faraday constant, and  $(1 + \delta_e)$  is the thermodynamic factor. Inserting the expressions for flux and current (Equation 12a and 12b respectively) into the PDEs and BCs gives

$$-\nabla \cdot (\sigma_e \nabla \phi_e) - \nabla \cdot (\sigma_e \theta \nabla \ln c_e) = 0 \quad \text{in } \Omega_e, \quad (14a)$$

$$\mathbf{n} \cdot (-\sigma_e \nabla \phi_e - \sigma_e \theta \nabla \ln c_e) = -j_n \quad \text{at } \partial\Omega_{n,e}, \quad (14b)$$

for charge conservation and

$$\partial_t c_e + \nabla \cdot \left( -D_e \nabla c_e + \frac{t_+^0}{F} \mathbf{i}_e \right) = 0 \quad \text{in } \Omega_e, \quad (15a)$$

$$-\mathbf{n} \cdot \left( -D_e \nabla c_e + \frac{t_+^0}{F} \mathbf{i}_e \right) = F^{-1} j_n \quad \text{at } \partial\Omega_{n,e}, \quad (15b)$$

for mass conservation. Equation 15 can be manipulated by using Equation 10 and assuming that gradients of the transference number are negligible,  $\nabla t_+^0 \approx 0$ . This is generally a reasonable assumption given that  $t_+^0$  usually does not vary significantly with  $c_e$  [29, 31]. We now have,

$$\partial_t c_e = \nabla \cdot (D_e \nabla c_e) \quad \text{in } \Omega_e, \quad (16a)$$

$$\mathbf{n} \cdot (D_e \nabla c_e) = F^{-1} (1 - t_+^0) j_n \quad \text{at } \partial\Omega_{n,e}. \quad (16b)$$

### 2.2.2 Averaging and Closure

The process for applying the VAT to the microscale equations is summarised in Figure 2. As per the first step, volume-averaged forms of the PDEs are written by applying the average operator (Equation 4). For charge conservation (Equation 14a) this gives,

$$-\langle \nabla \cdot (\sigma_e \nabla \phi_e) \rangle^e - \langle \nabla \cdot (\sigma_e \theta \nabla \ln c_e) \rangle^e = 0, \quad (17)$$

whilst for mass conservation (Equation 16a),

$$\partial_t \langle c_e \rangle^e = \langle \nabla \cdot (D_e \nabla c_e) \rangle^e. \quad (18)$$

Since the limits of the integration in the averaging operator are independent of time, integration and differentiation can be exchanged in the first term, and thus  $\langle \partial_t c_e \rangle^e = \partial_t \langle c_e \rangle^e$ . Now, following the second step, the averaged PDEs (Equations 17 and 18) will be subtracted from their microscale counterparts (Equations 14a and 16a). For charge conservation,

$$-\nabla \cdot (\sigma_e \nabla \phi_e) + \langle \nabla \cdot (\sigma_e \nabla \phi_e) \rangle^e - \nabla \cdot (\sigma_e \theta \nabla \ln c_e) + \langle \nabla \cdot (\sigma_e \theta \nabla \ln c_e) \rangle^e = 0, \quad (19)$$

whilst for mass conservation,

$$\partial_t c_e - \partial_t \langle c_e \rangle^e = \nabla \cdot (D_e \nabla c_e) - \langle \nabla \cdot (D_e \nabla c_e) \rangle^e. \quad (20)$$

To develop a closure problem, the microscale variables are decomposed into an average and a deviation from the average (Step 3). The decomposition for  $\phi_e$  was introduced above (Equation 7b), and that for  $c_e$  is,

$$c_e = \langle c_e \rangle^e + \tilde{c}_e. \quad (21)$$

The logarithm in Equation 19 introduces a non-linearity, which complicates the eventual development of a closure problem. For this reason,  $\nabla \ln c_e$  will be written as  $\frac{1}{c_e} \nabla c_e$ , and the  $c_e$  in the denominator approximated by  $\langle c_e \rangle^e$  – other articles in the literature [32, 33] also employ this assumption when upscaling electrolyte transport equations,

$$-\nabla \cdot (\sigma_e \nabla \phi_e) + \langle \nabla \cdot (\sigma_e \nabla \phi_e) \rangle^e - \nabla \cdot (\sigma_e \theta^* \nabla c_e) + \langle \nabla \cdot (\sigma_e \theta^* \nabla c_e) \rangle^e = 0, \quad (22)$$

where

$$\theta^* = \frac{\theta}{\langle c_e \rangle^e}. \quad (23)$$

Inserting the decompositions (Equation 7b and 21) into the charge conservation PDE (Equation 22) gives

$$\begin{aligned} & -\nabla \cdot (\sigma_e \nabla \langle \phi_e \rangle^e) - \nabla \cdot (\sigma_e \nabla \tilde{\phi}_e) + \langle \nabla \cdot (\sigma_e \nabla \langle \phi_e \rangle^e) \rangle^e + \langle \nabla \cdot (\sigma_e \nabla \tilde{\phi}_e) \rangle^e \\ & - \nabla \cdot (\sigma_e \theta^* \nabla \langle c_e \rangle^e) - \nabla \cdot (\sigma_e \theta^* \nabla \tilde{c}_e) + \langle \nabla \cdot (\sigma_e \theta^* \nabla \langle c_e \rangle^e) \rangle^e + \langle \nabla \cdot (\sigma_e \theta^* \nabla \tilde{c}_e) \rangle^e = 0, \end{aligned} \quad (24)$$

whilst for mass conservation (Equation 20),

$$\partial_t \tilde{c}_e = \nabla \cdot (D_e \nabla \langle c_e \rangle^e) + \nabla \cdot (D_e \nabla \tilde{c}_e) - \langle \nabla \cdot (D_e \nabla \langle c_e \rangle^e) \rangle^e - \langle \nabla \cdot (D_e \nabla \tilde{c}_e) \rangle^e. \quad (25)$$

Here, we introduce the assumptions that

$$\langle \nabla \cdot (\sigma_e \nabla \langle \phi_e \rangle^e) \rangle^e \approx \nabla \cdot (\sigma_e \nabla \langle \phi_e \rangle^e), \quad (26a)$$

$$\langle \nabla \cdot (\sigma_e \theta^* \nabla \langle c_e \rangle^e) \rangle^e \approx \nabla \cdot (\sigma_e \theta^* \nabla \langle c_e \rangle^e), \quad (26b)$$

$$\langle \nabla \cdot (D_e \nabla \langle c_e \rangle^e) \rangle^e \approx \nabla \cdot (D_e \nabla \langle c_e \rangle^e), \quad (26c)$$

since the averaged variables,  $\nabla \langle \phi_e \rangle^e$  and  $\nabla \langle c_e \rangle^e$  do not vary significantly over the averaging set,  $\mathcal{V}_e(\mathbf{x})$  – assumptions of this nature are discussed in depth in the VAT literature (see ‘Technical Note 8’ in Davit’s work [27] for example). We will also ignore variations in  $D_e$ ,  $\sigma_e$  and  $\theta^*$  over the scale of an averaging volume, thus allowing them to be removed from the divergence operators. This simplifies Equation 24 to

$$-\Delta \tilde{\phi}_e + \langle \Delta \tilde{\phi}_e \rangle^e - \theta^* \Delta \tilde{c}_e + \theta^* \langle \Delta \tilde{c}_e \rangle^e = 0, \quad (27)$$

whilst Equation 25 becomes,

$$\partial_t \tilde{c}_e = D_e \Delta \tilde{c}_e - \langle D_e \Delta \tilde{c}_e \rangle^e. \quad (28)$$

The quasi-steady state assumption [27] will be employed in Equation 28, thus  $\partial_t \tilde{c}_e \approx 0$ . Although this assumption is common in the VAT, the development could be advanced without it, which would result in a time dependent closure problem [19, 21, 27]. Equation 28 becomes

$$\Delta \tilde{c}_e = \langle \Delta \tilde{c}_e \rangle^e. \quad (29)$$

We will now shift our attention to the BCs. Continuing with Step 3, variable decompositions are inserted. For charge conservation (Equation 10b),

$$\mathbf{n} \cdot \left( -\sigma_e \nabla \langle \phi_e \rangle^e - \sigma_e \nabla \tilde{\phi}_e - \sigma_e \theta^* \nabla \langle c_e \rangle^e - \sigma_e \theta^* \nabla \tilde{c}_e \right) = -j_n, \quad (30)$$

and mass conservation (Equation 16b),

$$\mathbf{n} \cdot (D_e \nabla \langle c_e \rangle^e + D_e \nabla \tilde{c}_e) = F^{-1} (1 - t_+^0) j_n. \quad (31)$$

Here, two important assumptions are made – we will ignore the influence of the surface reaction,  $j_n$ , on both  $\tilde{\phi}_e$  and  $\tilde{c}_e$ . Such assumptions are also employed when deriving the ‘standard’ DFN using the VAT [32, 33], and can be justified on the basis that  $|j_n| \ll |\sigma_e \partial_x \langle \phi_e \rangle^e|$  and  $|F^{-1} (1 - t_+^0) j_n| \ll |D_e \partial_x \langle c_e \rangle^e|$  respectively (concentration and potential gradients are primarily along the  $x$  dimension). DNS reveal that these conditions often hold true for typical battery conditions (especially at lower C rates). Nonetheless, they are not mandatory, and a closure could be developed with the effect of  $j_n$  included. This is demonstrated in the SM (Section 4), where we show that  $j_n$  source terms are of secondary importance compared to those for  $\nabla \langle \phi_e \rangle^e$  in this application. The reader is also referred to Lenne’s work [33] for an alternative approach to modelling the surface reaction’s influence on  $\tilde{\phi}_e$  and  $\tilde{c}_e$ . For charge conservation, the BC is now

$$\mathbf{n} \cdot \left( -\nabla \langle \phi_e \rangle^e - \nabla \tilde{\phi}_e - \theta^* [\nabla \langle c_e \rangle^e + \nabla \tilde{c}_e] \right) = 0, \quad (32)$$

whilst for mass conservation,

$$\mathbf{n} \cdot (\nabla \langle c_e \rangle^e + \nabla \tilde{c}_e) = 0. \quad (33)$$

Notice that equations on  $\tilde{c}_e$  can be used to simplify those on  $\tilde{\phi}_e$ . By inserting Equation 29 into Equation 27 and Equation 33 into Equation 32 we arrive at

$$\Delta \tilde{\phi}_e = \langle \Delta \tilde{\phi}_e \rangle^e \quad \text{in } \Omega_e, \quad (34a)$$

$$\mathbf{n} \cdot \nabla \tilde{\phi}_e = -\mathbf{n} \cdot \nabla \langle \phi_e \rangle^e \quad \text{at } \partial\Omega_{n,e}. \quad (34b)$$

This system of equations reveals the macroscale source terms causing deviations in  $\phi_e$ , and leads us to define the following closure to relate  $\langle \phi_e \rangle^e$  and  $\phi_e$  (Step 4),

$$\tilde{\phi}_e = \mathbf{b} \cdot \nabla \langle \phi_e \rangle^e, \quad (35)$$

where  $\mathbf{b}(x, y, z) = [b_x, b_y, b_z]^T$  is a vector of closure (or mapping) variables. Although a more complex closure expression could be used, it will reduce to Equation 35 due to the assumptions already introduced (this is demonstrated in Section 5 of the SM). Since we are primarily interested in  $\tilde{\phi}_e$ , the treatment of  $\tilde{c}_e$  has also been relegated to the SM. To obtain a boundary value problem (BVP) to solve for  $\mathbf{b}$  (Step 5), Equation 35 is inserted into Equation 34,

$$\Delta \mathbf{b} = \langle \Delta \mathbf{b} \rangle^e \quad \text{in } \Omega_e, \quad (36a)$$

$$\mathbf{n} \cdot \nabla \mathbf{b} = -\mathbf{n} \quad \text{at } \partial\Omega_{n,e}, \quad (36b)$$

where higher order derivatives of spatially-smoothed variables have been neglected, arguing that  $\Delta(\mathbf{b} \cdot \nabla \langle \phi_e \rangle^e) \approx \Delta \mathbf{b} \cdot \nabla \langle \phi_e \rangle^e$  (this is standard in the VAT [27, 34]). We will expand the notation of  $\langle \Delta \mathbf{b} \rangle^e$ ,

$$\langle \Delta \mathbf{b} \rangle^e = \frac{1}{V_e} \int_{V_e} (\nabla \cdot \nabla \mathbf{b}) dV, \quad (37)$$

and using the divergence theorem, we find

$$\langle \Delta \mathbf{b} \rangle^e = \frac{1}{V_e} \int_{\mathcal{A}_{\text{tot}}} (\mathbf{n} \cdot \nabla \mathbf{b}) dA, \quad (38)$$

where  $\mathcal{A}_{\text{tot}}$  is the entire closed surface surrounding  $V_e$ . As well as interfacing with AM, the averaging volume  $V_e$  may also meet external boundaries (the current collector, outer walls, and a lithium foil if a half cell is being modelled). However, the BC at all external boundaries is  $\mathbf{n} \cdot \nabla \mathbf{b} = 0$  (see the SM for a further discussion), and Equation 38 therefore reduces to

$$\langle \Delta \mathbf{b} \rangle^e = -\frac{1}{V_e} \int_{\mathcal{A}} \mathbf{n} dA, \quad (39)$$

where  $\mathcal{A}(\mathbf{x}) \subset \partial\Omega_{n,e}$  is the internal boundary between electrolyte and AM in the averaging set. In the case of a battery, gradients in  $\langle \phi_e \rangle^e$  are predominantly along the thickness of the cell, the  $x$  dimension. We will therefore neglect  $\partial_y \langle \phi_e \rangle^e$  and  $\partial_z \langle \phi_e \rangle^e$ , leaving only the  $b_x$  component of the  $\mathbf{b}$  vector. The expression for  $\tilde{\phi}_e$  becomes

$$\tilde{\phi}_e = b_x \partial_x \langle \phi_e \rangle^e, \quad (40)$$

and the closure problem on  $b_x(x, y, z)$  is

$$\Delta b_x = -\frac{1}{V_e} \int_{\mathcal{A}} n_x dA \quad \text{in } \Omega_e, \quad (41a)$$

$$\mathbf{n} \cdot \nabla b_x = -n_x \quad \text{at } \partial\Omega_{n,e}. \quad (41b)$$

We have now arrived at a BVP describing the closure variable  $b_x$  throughout the entirety of the electrolyte within the electrode,  $\Omega_e$ . We will now turn our attention to the battery separator.

### 2.2.3 The Separator

In a real battery cell, the separator is a porous solid filled with electrolyte. To simplify the presentation of the closure problem, only the development within the electrode was shown. However, the problem can easily be extended to include the electrolyte within the separator region: in this case,  $\Omega_e = \Omega_{e,n} \cup \Omega_{e,s}$ , where  $\Omega_{e,n}$  and  $\Omega_{e,s}$  are the sets containing all electrolyte points in the negative electrode and separator respectively. The electrolyte transport PDEs (Equations 10a and 11a) are unchanged in the separator, and at the boundary between electrolyte and the inactive solid structure,  $\partial\Omega_{i,e}$  the BCs are,

$$\begin{aligned} \mathbf{n} \cdot \mathbf{N}_e &= 0 & \text{at } \partial\Omega_{i,e}, \\ \mathbf{n} \cdot \mathbf{i}_e &= 0 & \text{at } \partial\Omega_{i,e}. \end{aligned}$$

Following an identical procedure to the development above, the BC within the closure problem will be,

$$\mathbf{n} \cdot \nabla b_x = -n_x \quad \text{at } \partial\Omega_{i,e}.$$

Including the real structure of the separator is challenging from a modelling perspective due to the disparity in length scales – a typical separator pore diameter is  $\sim 50$  nm [35], requiring a mesh far finer than that used for the electrode. Consequently, the separator is often modelled as a homogenised

medium [14, 36, 37], in which transport PDEs are unchanged, but effective transport properties are introduced to account for the reduced transport efficiency caused by the solid structure.

Modelling a homogenised separator complicates the theory developed above significantly: the closure must be solved in  $\Omega_{e,n}$  and  $\Omega_{e,s}$  separately, with the two problems coupled by BCs at the separator-electrode interface. We argue that this additional complexity is not necessary for the objectives of this work – whilst using a homogenised separator is reasonable for many applications, it is less appropriate for studying lithium-plating onset, where behaviour at the interface between AM and the separator electrolyte plays a critical role (discussed further below). Furthermore, in its current form, the closure problem is able to model a non-homogenised separator, which is of course the situation found in a real battery cell. The validation performed below is split into two sections. Firstly the separator region is treated as pure electrolyte, as a ‘proof of concept’ for the theory developed. Secondly, a real separator structure is modelled.

#### 2.2.4 Contrast with the Conventional VAT

Conventionally in the VAT, a representative unit cell would be defined, and the closure problem would be solved only on this unit cell [21, 27, 38]. In this work, we instead solve the closure problem across the entirety of  $\Omega_e$ . This gives rise to important nuances compared to the conventional VAT – a schematic to aid this discussion is shown in Figure 3. Firstly, in the conventional VAT, periodic conditions are usually imposed at the external boundaries of the unit cell,  $\partial\Omega_{\text{ext}}^{\text{uc}}$ . Since we solve for  $b_x$  across the entire microscale domain, BCs must be determined at the external boundaries (the current collector, outer walls, and lithium foil). As mentioned above, these unanimously take the form  $\mathbf{n} \cdot \nabla b_x = 0$ , and are discussed in the SM. Notice that the BVP on  $b_x$  is determined up to a constant, and thus another condition is required for a unique solution. In the conventional VAT, this additional constraint is typically

$$\int_{\Omega_e^{\text{uc}}} b_x dV = 0, \quad (43)$$

where  $\Omega_e^{\text{uc}}$  is the set containing all electrolyte points within the unit cell. A different approach is employed here, which produces more useful results for our application. Since the interface between the separator and the negative electrode is known to pose the highest risk of lithium plating [8], we are primarily interested in electrolyte potential variations here (rather than at other locations within the electrode). Our end goal is to take an averaged value of electrolyte potential,  $\langle \phi_e \rangle^e$ , and reconstruct microscale values of electrolyte potential using our knowledge of deviations from the average,  $\tilde{\phi}_e$ . We therefore desire the following property,

$$\int_{\mathcal{A}_{yz}} \tilde{\phi}_e dA = 0, \quad (44)$$

where  $\mathcal{A}_{yz} \subset \Omega_e$  (see Figure 3) is the  $yz$  cross section at the interface between the separator and the negative electrode (the set containing all points within  $\Omega_{e,s}$  that interface with  $\Omega_{e,n}$ ). Equation 44 leads to,

$$\int_{\mathcal{A}_{yz}} b_x dA = 0. \quad (45)$$

A unique solution can now be obtained for  $b_x$ . If we are interested in reconstructing overpotential at some other point in the electrode, we would simply define the averaging slice ( $\mathcal{A}_{yz}$ ) at a different  $x$  position. Such an approach therefore gives rise to a family of problems for  $b_x$ : the PDE (Equation 41) is solved only *once*, which specifies  $b_x$  up to an integration constant, but the unique values of  $b_x$  are determined by setting  $\mathcal{A}_{yz}$  at the  $x$  position of interest.

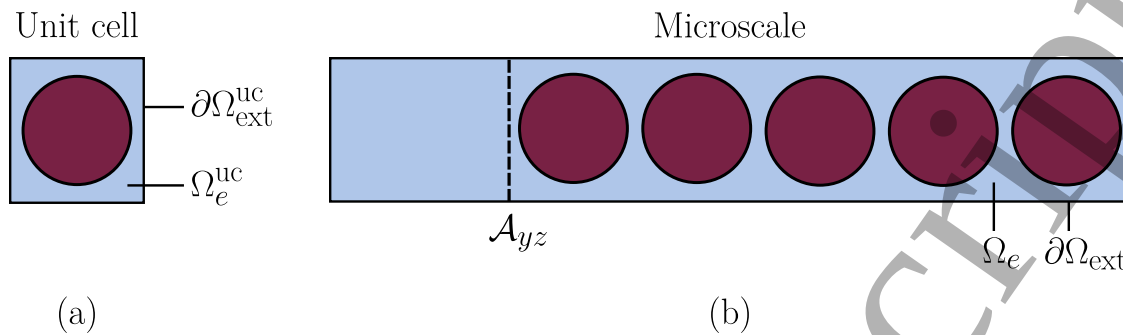


Figure 3: (a) Illustrating the definition of a unit cell used in the conventional VAT and (b) the approach used in our framework.

### 2.2.5 Reconstruction Using the Closure

Using the solution to the closure problem, the electrolyte potential and lithium-plating overpotential can be estimated. Inserting Equation 40 into Equation 7b gives

$$\phi_e = \langle \phi_e \rangle^e + b_x \partial_x \langle \phi_e \rangle^e. \quad (46)$$

This is a crucial development – with knowledge of  $b_x$  and solutions to an upscaled model,  $\phi_e$  can be reconstructed everywhere in the microstructure's electrolyte domain *without* solving the full microscale model. Although the closure itself is solved on the microstructure and thus relatively computationally expensive, this must only be done once to characterise an electrode geometry; the solution is independent of material properties and operating conditions. With the reconstruction of  $\phi_e$ , we can now reach our destination: an estimate of the lithium-plating overpotential everywhere on  $\partial\Omega_{n,e}$  (denoted  $\eta_{\text{pl}}^r$ , where the superscript  $r$  stands for 'reconstructed'). Inserting Equation 40 into Equation 9 gives,

$$\eta_{\text{pl}}^r = \langle \phi_n \rangle^n - \langle \phi_e \rangle^e - b_x \partial_x \langle \phi_e \rangle^e. \quad (47)$$

Finally, an improved estimate of the minimum plating overpotential,  $\eta_{\text{pl,min}}^r$ , can be obtained, which will be used to identify when to terminate a charge (discussed further below),

$$\eta_{\text{pl,min}}^r = \min (\langle \phi_n \rangle^n - \langle \phi_e \rangle^e - b_x \partial_x \langle \phi_e \rangle^e). \quad (48)$$

## 3 Validation Methodology

We validate the theory developed above in two sections. The philosophy and methodology of the first section follow a very similar structure to Teel's work [8]. We generate several artificial graphite electrodes which possess identical macroscale properties (volume fractions and mean particle radius). However, the electrodes have different microstructures, and thus lithium plating becomes favourable at different times and locations. The first key difference between this section and Teel's work is that we parameterise macroscale models – the DFN and the DC formulation. The second key difference is that we evaluate the ability of the closure developed above for identifying *local* lithium-plating onset. The final key difference is that we test the correction on  $\eta_{\text{pl,min}}^r$  (Equation 48) for predicting when lithium plating first becomes thermodynamically favourable.

In the first section, the separator region is modelled as pure electrolyte. This demonstrates the prospect of using closure to predict lithium-plating onset, whilst also simulating a (relatively) large portion of the electrode. In the second section, we simulate a case with a real separator structure. This showcases the closure for studying lithium-plating onset at the interface with a real separator, but simulates a (relatively) small portion of the electrode, due to the mesh resolution constraints discussed above.

Table 1: MATBOX input parameters.

| Parameter                                         | Value(s)                   |
|---------------------------------------------------|----------------------------|
| Electrode side length [voxels]                    | 137                        |
| Voxel side length [ $\mu\text{m}$ ]               | 0.476                      |
| Mean particle diameter [voxels]                   | 21                         |
| Targeted AM volume fraction                       | 0.66                       |
| Standard deviation [voxels]                       | 4                          |
| Diameter distribution [voxels]                    | 15, 17, 19, 21, 23, 25, 27 |
| Diameter distribution weighting [%]               | 14, 14, 14, 16, 14, 14, 14 |
| Overlapping parameter                             | 0.5                        |
| Minimum fraction of particle volume conserved [%] | 90                         |



Figure 4: The half cells for Electrode U (a) and Electrode R (b), with the electrolyte in light blue, AM in dark blue, the inactive separator solid in red, and the current collector in grey.

### 3.1 Part I Microstructure Generation

For Part I, we used the MATBOX application [39] to generate 5 artificial electrodes composed of overlapping spheres. The parameters used to define the electrodes (shown in Table 1) are identical to those used by Teel, with the exception of the standard deviation (Teel used a value of  $0.952 \mu\text{m}$ , whilst we choose  $1.904 \mu\text{m}$  to accentuate the differences between the electrode microstructures). The electrodes are cubic with a side length of  $65.212 \mu\text{m}$ . Identically to Teel, we first generate a ‘uniform’ electrode, dubbed Electrode U, for which only a single diameter is input. We then generate 4 electrodes with the same mean radius, but a standard deviation of 4 voxels ( $1.9 \mu\text{m}$ ). Due to the random nature of the generation algorithm, each run will output a different microstructure, despite macroscale properties being consistent.

We use half cells to validate the theory developed, to isolate the behaviour of the negative electrode – the half-cell mesh for the Electrode U is shown in Figure 4, and a full schematic is given in the SM. We refer to the working electrode as ‘negative.’ This is not technically accurate, since in a half cell the electrode would in fact be positive relative to a lithium foil. Nonetheless we maintain this convention to avoid confusion, as we are interested in the negative electrode of a full cell.

3.2 Part II Microstructure Generation

To build a half cell with the separator microstructure included, an SEM image of a Celgard® PP1615 separator [40] was used. The original dataset has a voxel size of 10 nm. Since this is much finer than the typical resolution of an electrode mesh, the binary image was coarsened using a block-wise majority rule. The original voxel grid was partitioned into non-overlapping cubic blocks of size 5×5×5 voxels. Each block was replaced by a single voxel whose value was determined by the phase occupying the majority of the block volume. This procedure preserves the binary nature of the image while reducing the spatial resolution and maintaining the dominant phase connectivity at the coarser scale. Finally, a 128×96×137 voxel subsection was taken from the course image.

The electrode used for Part II is dubbed ‘Electrode R’, indicating that this case uses the ‘real’ separator. Electrode R was built from Electrode U: a subsection was taken (137×96×137 voxels) and the voxel side length was rescaled to be 50 nm rather than 0.476 μm to match the voxel size of the separator dataset. The mean particle diameter therefore becomes 1.05 μm – although this is relatively small, it is not outside the range of plausibility for graphite particles [41, 42]. Likewise, the overall dimensions of the electrode are 6.85×4.8×6.85 μm. Clearly, this is a small portion of an electrode. We reiterate that the purpose of Part II is to complement Part I (which simulates a much larger electrode volume) by demonstrating the closure’s ability to reconstruct lithium-plating potential at the interface between an electrode and a real separator. The resulting half cell when Electrode R is combined with the separator structure is shown in Figure 4.

3.3 Model Parameterisation

The graphite material properties used in all models are shown in Table 2. To be concise, the model equations for the DFN and DC are not provided here: these are included in the SM, and discussed in depth in our original publication of the DC model. For both upscaled models, the majority of parameters are identical to those used in the microscale model, facilitating a direct comparison. However, certain parameters do not exist in the microscale model: phase volume fractions, transport efficiencies, specific surface area, particle radius (for DFN), and the closure variable (for DC).

Each of these were set exactly as in our previous work [18, 19]. Phase volume fractions ( $\varepsilon_e$  and  $\varepsilon_n$ ) were calculated directly from the electrode images, by summing the number of voxels of each phase. Similarly, the number of voxel faces at the phase interface were counted to find the specific surface area,  $a_n$ . Transport efficiencies for each phase ( $\mathfrak{B}_e$  and  $\mathfrak{B}_n$ ) were also calculated directly from the microstructure, following the method outlined by Kehrwald et al. [43]: this was done with the in-house code used to solve the DNS (discussed below), which has been validated against Taufactor [44] for this application.

When generating an electrode with MATBOX, a label map is also produced identifying particle boundaries. This was used in the calculation of the closure variable parameter for DC,  $\langle s \rangle_A$ ; we solved the closure problem on the electrode microstructure using the tool developed in our previous work [45]. For the DFN, the radius parameter,  $R_n$ , is set as the mean radius value that was input to MATBOX.

3.4 Simulation Methods

To create a ‘fast charge’ situation, a 2C charge was simulated for each case, which was terminated when lithium plating becomes favourable at any point in the electrode ( $\eta_{pl,min} \leq 0$ ). C-rate was determined by assuming a stoichiometry of 0.0279 at 0% state-of-charge (SOC), and 0.9014 at 100% SOC, as measured by Chen et al. [28].

To obtain the DNS results, used as a reference to evaluate both macroscale models, an in-house code was used to solve the microscale model equations (as outlined in the SM). This solver employs a finite-volume scheme and a voxel-based mesh, and has been validated against BEST [47, 48] and COMSOL Multiphysics [49]. Both the DFN and DC were solved using PyBaMM [50] (see SM for numerical details).

Table 2: The material properties and initial conditions used.

| Property Name                                                                          | Symbol           | Value                 |      |
|----------------------------------------------------------------------------------------|------------------|-----------------------|------|
| Thermodynamic factor                                                                   | $(1 + \delta_e)$ | 1                     | [29] |
| Transference number                                                                    | $t_+^0$          | 0.4                   | [29] |
| Electrolyte conductivity [ $\text{Sm}^{-1}$ ]                                          | $\sigma_e$       | 1                     | [29] |
| Electrolyte diffusivity [ $\text{m}^2\text{s}^{-1}$ ]                                  | $D_e$            | $1 \times 10^{-10}$   | [29] |
| Initial electrolyte concentration [ $\text{molm}^{-3}$ ]                               | $c_{e,0}$        | 1000                  | [46] |
| Initial AM concentration [ $\text{molm}^{-3}$ ]                                        | $c_{n,0}$        | 924.41                | [28] |
| Temperature [K]                                                                        | $T$              | 298.15                | n/a  |
| Maximum AM concentration [ $\text{molm}^{-3}$ ]                                        | $c_{n,\max}$     | 33133                 | [28] |
| AM conductivity [ $\text{Sm}^{-1}$ ]                                                   | $\sigma_n$       | 100                   | [8]  |
| AM diffusivity [ $\text{m}^2\text{s}^{-1}$ ]                                           | $D_n$            | $3.3 \times 10^{-14}$ | [28] |
| Negative electrode rate constant [ $\text{Am}^{-2}(\text{m}^3\text{mol}^{-1})^{1.5}$ ] | $Fk_n$           | $6.48 \times 10^{-7}$ | [28] |
| Exchange-current density for lithium foil [ $\text{Am}^{-2}$ ]                         | $j_{li,0}$       | 10                    | [33] |

The closure problem given in Equation 41 was solved using COMSOL Multiphysics [49]. For each geometry, an STL file was generated to represent the MATBOX output (TIF file) exactly. This was then used to build the mesh in COMSOL – no smoothing was performed, to ensure that the resulting geometry accurately represents the voxel-based mesh used in the DNS. A finite-element mesh was generated using COMSOL’s mesh generation tool with the ‘normal’ setting for element size. Finally, an iterative GMRES solver was used.

## 4 Results I: Several Electrodes with Electrolyte Separators

### 4.1 Local Plating Onset

As commonly observed [8, 51, 52], the area most at risk of lithium plating (where  $\eta_{\text{pl}}$  is lowest) is the interface of the negative electrode with the electrolyte in the separator region – Figure 5a shows the solution for the closure variable,  $b_x$ , across this interface in Electrode U. In combination with an upscaled model providing  $\langle \phi_e \rangle^e$  (the DC model is used here), the local electrolyte potential  $\phi_e$  may be reconstructed with Equation 46, as shown in Figure 5b. The resulting reconstruction of the plating overpotential,  $\eta_{\text{pl}}^r$ , is shown in Figure 5d.

By comparing  $\eta_{\text{pl}}^r$  (Figure 5d) with the DNS reference,  $\eta_{\text{pl}}$  (Figure 5c), **it is clear that the reconstruction from the closure predicts the regions prone to plating with high fidelity.** Most importantly, the location where lithium-plating first becomes thermodynamically favourable is correctly identified by the reconstructed field (see the upper-central region of the interface). Likewise, other areas at high risk of plating are pinpointed, and the zones of lowest risk are clearly seen. **The absolute values predicted by  $\eta_{\text{pl}}^r$  are less reliable.** Specifically, the range of  $\eta_{\text{pl}}^r$  is considerably larger than  $\eta_{\text{pl}}$ . This is due to the difficulty of predicting  $\partial_x \langle \phi_e \rangle^e$  accurately at the separator-electrode interface – since the two regions have different transport efficiencies, the potential profile  $\langle \phi_e \rangle^e$  can experience a relatively sudden change across this interface. The macroscale models overestimate  $\partial_x \langle \phi_e \rangle^e$  compared to the DNS (in this case), thus resulting in an overestimation in the range of  $\eta_{\text{pl}}^r$  (see Table 5 in the SM for  $\partial_x \langle \phi_e \rangle^e$  values).

A comparison of  $\eta_{\text{pl}}$  and  $\eta_{\text{pl}}^r$  is shown for the other four electrodes in Figure 7. Owing to the

stochastic nature of the MATBOX generation algorithm, each electrode exhibits a distinct interfacial geometry between the separator and the AM. Despite this variability, Figure 7 reinforces the conclusions made above: that  $\eta_{\text{pl}}^r$  is able to faithfully identify the local lithium-plating ‘hot spots’, despite its overestimation in the range of the real plating overpotential.

Besides the difficulty in estimating  $\partial_x \langle \phi_e \rangle^e$ , errors can also be attributed to the simplifications made in the closure development. Most notably, the influence of the reaction source term at  $\partial \Omega_{\eta,e}$  was neglected (see Equation 32). **Nonetheless, we explore the magnitude of these source terms in the SM (Section 4), and find that even at 2C they are two orders of magnitude smaller than the potential-gradient source terms.** In addition, the deviations in solid potential,  $\phi_n$ , were neglected in our development (see Equation 9). These could also be included, in which case a closure problem analogous to Equation 41 must be solved in the AM phase. While such extensions would improve the accuracy of  $\eta_{\text{pl}}^r$  to some degree, they would also substantially increase the computational and implementation complexity. Given their expected secondary influence relative to the dominant source terms already included, we do not pursue them here. Moreover, regarding the deviations in solid potential, the DNS results confirm that the assumption  $\tilde{\phi}_n \ll \tilde{\phi}_e$  is very well justified in these cases, and the correction gained by considering  $\tilde{\phi}_n$  is in the order of  $\mu\text{V}$ .

The primary application we envisage for the framework developed here is as a rapid and efficient means to predict *local* lithium-plating risk. Such an approach is particularly attractive for electrode design, as candidate microstructures could be assessed efficiently without resorting to computationally intensive DNS. To give an order-of-magnitude comparison, the closure problem took 2 minutes to solve, whilst the DNS took 45 minutes. Because different solvers and computers were used, the solution times are not directly comparable. However, the DNS was executed on a high-performance computing cluster with an efficiently parallelised implementation; we expect that applying the same approach to the closure problem would further reduce its computational cost. Although this time saving is considerable, the framework becomes exponentially more advantageous when a range of material properties or charging conditions are explored. One must solve a new DNS case for *every* parameter set. The closure problem, on the other hand, is solved *once* per geometry – creating a new  $\eta_{\text{pl}}^r$  field for different material properties or charging conditions requires  $\sim 0.1\text{s}$  (the time required to solve the macroscale model [18]). Finally, the closure problem’s independence from material properties and charging conditions gives the interesting perspective of understanding lithium-plating risk purely as a function of electrode geometry.

## 4.2 Minimum Plating Overpotential

One of the key results of Teel’s work is the plot of minimum lithium-plating overpotential within each electrode as a function of time (Figure 4 in their publication [8]). We produce similar figures in this section, but with DFN and DC included (Teel’s study only considered DNS). Further, the upscaled models both with (Equation 48) and without (Equation 6) the correction for minimum lithium-plating overpotential are studied.

The results for Electrode U are shown in Figure 6. For both upscaled models, the correction offered by the closure problem significantly improves their estimation of  $\eta_{\text{pl,min}}^r$ . Without the correction, the upscaled models considerably underestimate the likeliness of lithium-plating: for the DFN, the onset of lithium plating is predicted 68s late compared to the DNS, whilst this is 30s for DC. Including the correction from the closure problem reduces this value to 41s and 2s respectively. In this regard, DC is more accurate than DFN. This is due to the improved estimate of AM surface concentration in DC, which enhances the accuracy of  $\phi_n$  (see the original DC publication for further discussion).

The results for Electrodes 1, 2, 3, and 4 are shown in Figure 8. These demonstrate that the observations from Electrode U hold for different microstructures: the upscaled models underestimate plating risk, the correction from the closure improves upscaled model accuracy, and the DC model is more reliable than the DFN for predicting lithium-plating onset. The errors that remain in the upscaled models (after the correction from the closure has been applied) can be attributed to the sources discussed above: inaccuracy in  $\partial_x \langle \phi_e \rangle^e$  and simplifications in the closure development.

Table 3: The minimum values of the closure variable,  $b_x$ , on the separator interface.

| Electrode | Minimum $b_x$ [ $\mu\text{m}$ ] |
|-----------|---------------------------------|
| U         | -2.462                          |
| 1         | -2.952                          |
| 2         | -2.416                          |
| 3         | -3.559                          |
| 4         | -3.761                          |
| R         | -4.300                          |

Table 4: The time at which lithium-plating first became thermodynamically favourable.

| Model         | Time [s] |     |     |     |     |      |
|---------------|----------|-----|-----|-----|-----|------|
|               | U        | 1   | 2   | 3   | 4   | R    |
| DNS           | 789      | 828 | 802 | 815 | 801 | 1994 |
| DFN           | 857      | 914 | 885 | 905 | 893 | 1993 |
| DC            | 819      | 877 | 846 | 867 | 854 | 1995 |
| DFN corrected | 830      | 887 | 861 | 875 | 860 | 1993 |
| DC corrected  | 791      | 849 | 820 | 836 | 820 | 1994 |

One potential criticism of our findings is that in general, there is some uncertainty in the parameterisation of upscaled models, thus different parameter sets could be used in which case the conclusions above may not hold. Since we are comparing with DNS rather than experimental data, we argue that there is in fact minimal uncertainty in our parameterisation. The majority of the parameters required for the upscaled models are taken directly from the DNS parameter set (material properties, rate constants, etc.). The remaining parameters are calculated from the electrode images, and are set identically for the DFN and DC to allow fair comparison (volume fractions, transport efficiencies, and surface area). Normally, the primary area where ambiguity would exist is the definition of particle radius for the DFN; this would affect the DFN's predictions in Figures 6 and 8. However, this is not the case here, since the electrodes have been generated from spheres with a prescribed size distribution, and thus mean particle radius is well defined.

An interesting extension of this section would be to test more 'realistic' electrodes. The agreement of 2s between DNS and corrected-DC models quoted above can be considered a best-case result, as it was for an electrode made of spherical, monodisperse particles (although the other electrodes in Table 4 are indeed polydisperse). We expect that the differences between the DC and DFN results will diverge as the electrode departs from this idealised case, as the DFN's assumption of isolated, spherical particles is violated, whilst the DC model can handle arbitrarily complex microstructures via its closure problem (see Ref. [20]).

## 5 Results II: A Real Separator

### 5.1 Local Plating Onset

As in the Results I Section, the area most at risk of lithium plating (where  $\eta_{\text{pl}}$  is lowest) is the interface of the AM in Electrode R with the electrolyte in the separator region. The electrolyte-filled pore space

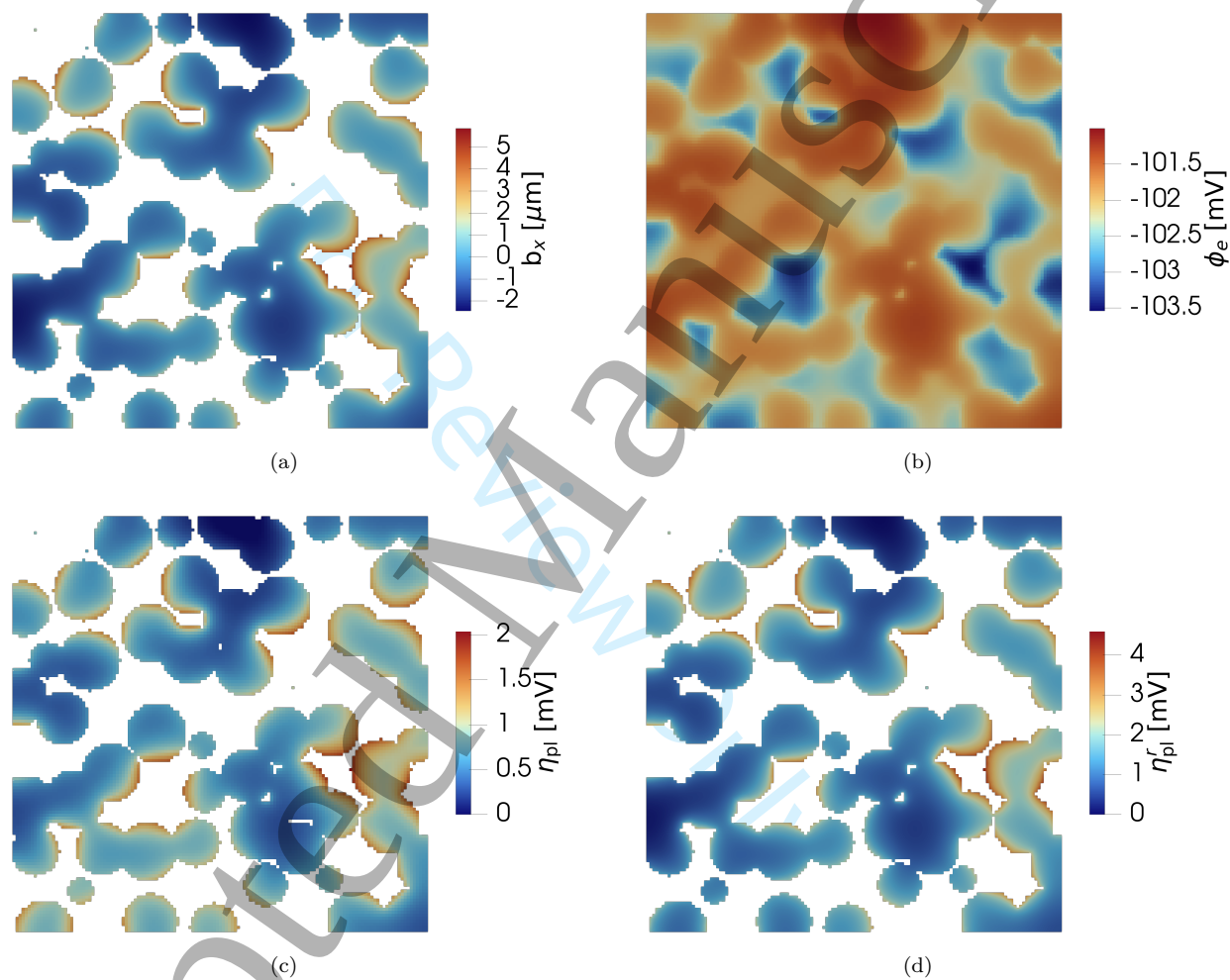


Figure 5: (a) The closure variable,  $b_x$ . (b) Electrolyte potential,  $\phi_e$ , reconstructed using  $b_x$ . (c) The lithium-plating overpotential from the DNS reference,  $\eta_{\text{pl}}$ . (d) The lithium-plating overpotential reconstructed from the closure problem,  $\eta_{\text{pl}}^r$ . All plots are evaluated at the separator interface of Electrode U at the point when the charge is terminated.

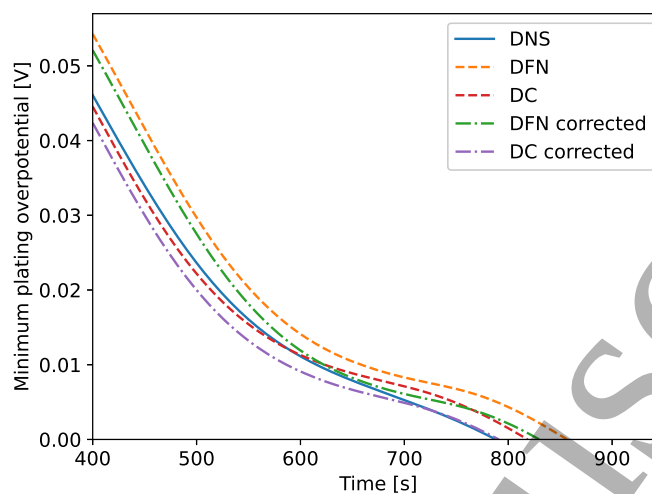


Figure 6: The minimum plating overpotential found at any point within the electrode against time for Electrode U, including the DNS reference alongside the DFN and DC models with and without correction.

of the separator is shown in Figure 9, with the relevant interface highlighted. The results for  $\eta_{pl}$  and  $\eta_{pl}^r$  are shown in Figure 10. As above, the reconstruction from the closure captures the critical features seen in the DNS reference with high fidelity. Most importantly, the area where plating risk is greatest (bottom left of Figure 10) is correctly identified. **In contrast to the Results I Section, where the absolute values of overpotential predicted by  $\eta_{pl}^r$  were inaccurate, the reconstruction is much more precise in this regard here.** This improvement arises (at least in part) from the explicit inclusion of the separator microstructure. It may also be partially caused by the lower electrolyte ohmic losses in Electrode R, discussed further below. **This supports the argument in the previous results section: errors in  $\eta_{pl}^r$  are primarily due to difficulties in predicting  $\partial_x \langle \phi_e \rangle^e$  with an upscaled model, rather than the absence of reaction-rate source terms in the closure. Here, the lower ohmic losses make  $\partial_x \langle \phi_e \rangle^e$  easier to predict, thus the absolute values of  $\eta_{pl}^r$  are more accurate.**

## 5.2 Minimum Plating Overpotential

As in the Results I Section, the minimum plating overpotential in Electrode R is shown in Figure 11. Two features are immediately apparent. Firstly, the differences between the various models are substantially smaller than in the previous results. Secondly, the time at which  $\eta_{pl,min}^r = 0$  occurs is much later than in the above cases (the electrode must actually be overcharged to reach  $\eta_{pl,min}^r = 0$ ). Both observations can be attributed to reduced electrolyte ohmic losses in Electrode R, since the half cell is significantly thinner; in other words, the average value of  $\partial_x \langle \phi_e \rangle^e$  is lower (see Table 5 in the SM). As per Equation 47, this in turn makes the range of  $\eta_{pl}^r$  smaller. Consequently, the condition  $\eta_{pl,min}^r = 0$  is reached later in the charge, and the differences between each model are less pronounced, as the correction offered by  $\eta_{pl,min}^r$  is not as significant.

The results in Figure 11 are of less interest than those in Figure 8, due to the dimensions of Electrode R being considerably smaller than a typical graphite electrode. The primary result of studying Electrode R is Figure 10 – confirming that the reconstruction from the closure is able to capture behaviour at the interface with a realistic separator structure.

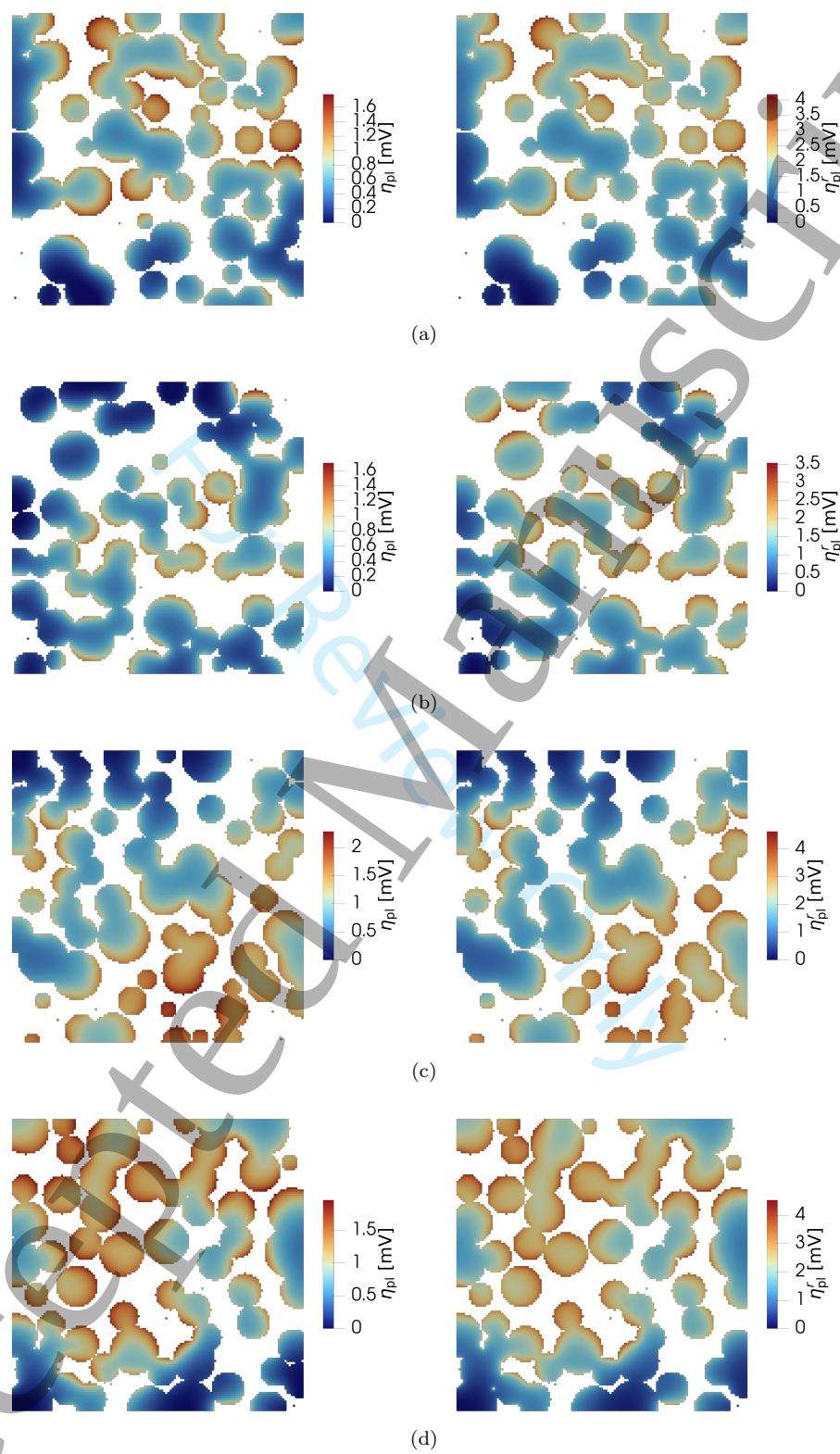


Figure 7: The DNS reference,  $\eta_{pl}$  (left), and the reconstruction from the closure,  $\eta_{pl}^r$  (right), at the end of charge for Electrodes 1 (a), 2 (b), 3 (c), and 4 (d). The slice shown is the interface between AM and electrolyte in the separator region.

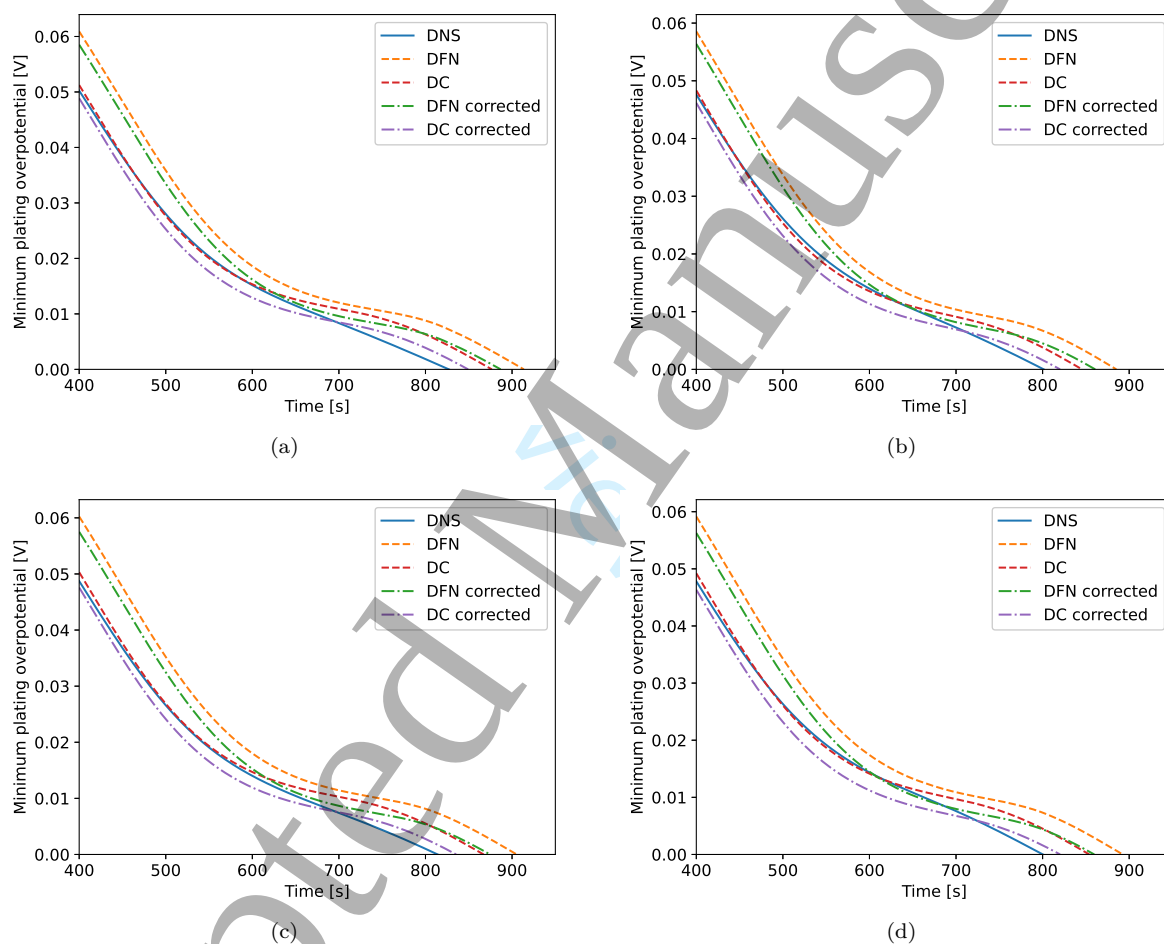


Figure 8: The minimum plating overpotential found at any point within the electrode against time for Electrodes 1 (a), 2 (b), 3 (c), and 4 (d).

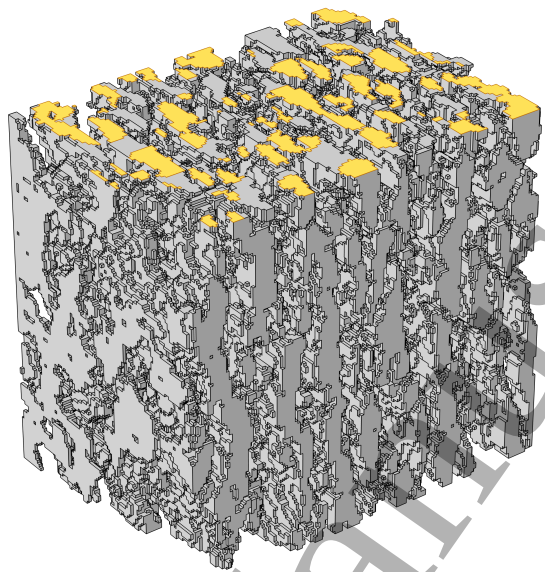


Figure 9: The electrolyte within the separator (grey), and its interface with the AM in Electrode R (yellow).

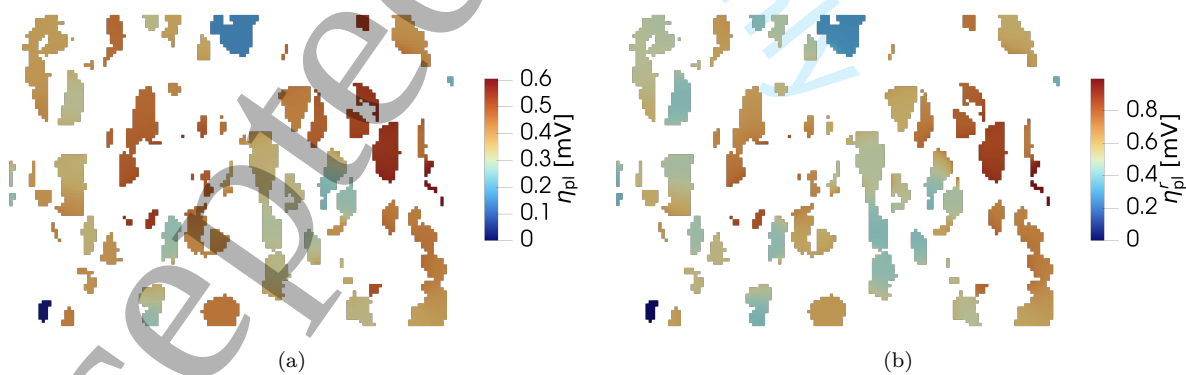


Figure 10: The DNS reference,  $\eta_{pl}$  (left), and the reconstruction from the closure,  $\eta_{pl}^r$  (right), at the interface between the separator electrolyte and the AM in Electrode R, taken at the end of the charge.

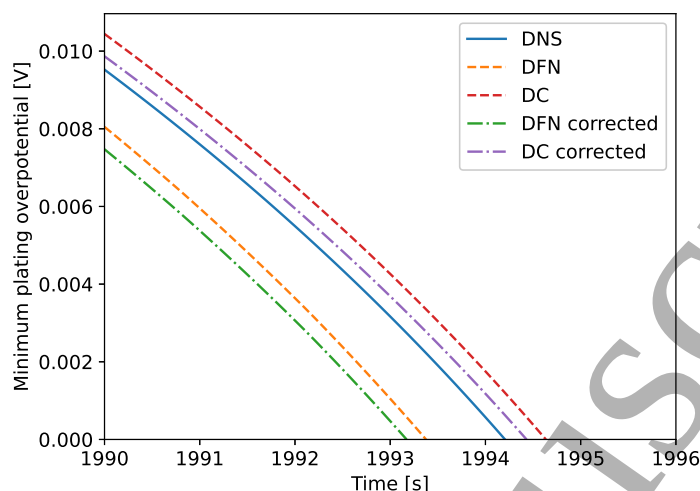


Figure 11: The minimum plating overpotential found at any point within the electrode against time for Electrode R.

## 6 Conclusions

In this work, we have extended the dual-continuum (DC) and Doyle-Fuller-Newman (DFN) modelling frameworks to enable the prediction of *local* lithium-plating onset within negative-electrode microstructures, while retaining the computational efficiency of upscaled approaches. Using a closure problem for electrolyte transport, we establish a direct link between averaged variables and microscale geometric features, thereby overcoming a key limitation of macroscale models, which inherently resolve only spatially-averaged quantities.

The closure-based reconstruction introduced here allows local fields of lithium-plating overpotential to be recovered from an upscaled model. Through comparison with direct numerical simulations (DNS), we demonstrate that this approach accurately identifies *where* lithium-plating risk is greatest. The closure was also used to propose a correction to the DC and DFN models, which improved their ability to predict *when* lithium plating first becomes thermodynamically favourable. Importantly, this framework provides a means to study lithium-plating onset within a microstructure without solving the computationally expensive DNS.

Some limitations of the present closure formulation remain. Whilst the reconstruction was able to accurately locate regions with high/low plating risk, the absolute values of lithium-plating overpotential were less reliable due to the upscaled models' imprecise predictions of electrolyte potential gradients. Assumptions employed in the closure development also introduce some error – e.g. ignoring the influence of reaction source terms at the electrolyte-active-material interface. While these effects could be incorporated through additional closure problems, doing so would increase the complexity of the reconstruction procedure, and their contributions were found to be secondary.

Overall, this work demonstrates that upscaled models with closure provide a viable and efficient route to predicting local lithium-plating behaviour in electrode microstructures, and that a detailed microscale model is not mandatory for this task. We envisage two primary applications for this framework: (i) in electrode design, to efficiently evaluate candidate microstructures without resorting to computationally intensive DNS; (ii) in upscaled models, to improve predictions of when lithium plating first becomes thermodynamically favourable.

A key open question for future work is: what geometrical features drive lithium-plating risk? Since the closure problem is purely a function of geometry, it offers a means to investigate this, with direct implications for electrode design. This study represents only a first step in exploiting the potential of closure-based approaches in lithium-ion battery modelling. Beyond lithium plating,

the framework developed here is readily extendable to other degradation mechanisms governed by microscale transport heterogeneity. In the present work, we focus exclusively on the onset of lithium plating; however, extending the approach to model the evolution of plating over a battery's lifetime would further amplify its advantages. In such scenarios, the need to simulate a large number of charge-discharge cycles renders DNS prohibitively expensive, whereas upscaled models with closure remain computationally tractable.

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